

TinyAD: Lightweight Anomaly Detection for Vibration Sensors on Microcontrollers

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Abstract

Predictive maintenance relies on detecting abnormal vibration in rotating machinery, yet most anomaly detectors are too large to run on the microcontrollers attached to the sensors. We present TinyAD, an autoencoder with depthwise-separable convolutions and 8-bit quantisation that occupies 46 kB of flash and processes a one-second window in 9 ms on a Cortex-M4 at 80 MHz. On three public bearing datasets TinyAD reaches an average AUROC of 0.962, within 0.011 of a 40-times larger model, and reduces radio traffic by 98% because only anomaly scores are transmitted.

1 Introduction

Industrial sites deploy thousands of battery-powered vibration sensors. Streaming raw data to the cloud drains batteries within months. On-device inference would extend battery life, but memory budgets of 64–256 kB rule out the recurrent and transformer models that dominate recent benchmarks.

Our contributions are: (i) an architecture search restricted to operations supported by common embedded inference runtimes; (ii) a calibration procedure that sets alarm thresholds per machine from 10 minutes of normal operation; and (iii) an evaluation on hardware, including energy measurements.

2 Related work

Spectral-feature methods with one-class classifiers are compact but require hand-tuned frequency bands. Deep autoencoders and isolation forests have been compressed by pruning, but rarely evaluated on device.

3 Method

Input windows of 1,024 samples are transformed by a fixed 64-band filter bank. The encoder has four depthwise-separable blocks; the decoder mirrors it. Reconstruction error, averaged over bands, is the anomaly score. We quantise weights and activations to 8 bits after training.

4 Experiments

Model	Flash (kB)	Latency (ms)	Mean AUROC
Isolation forest (spectral features)	120	15	0.901
Dense autoencoder	310	22	0.948
LSTM autoencoder (reference, server)	1,840	n/a	0.973
TinyAD (ours)	46	9	0.962

Energy per inference was 0.31 mJ, so a 2,400 mAh cell lasts about 4.3 years at one inference per minute.

5 Conclusion

Careful architecture choices make accurate anomaly detection feasible on the sensor itself.

References

- [1] J. Smith and R. Patel. Bearing fault datasets for benchmarking. Example Mechanical Systems, 2022.
- [2] L. Chen et al. Quantised autoencoders at the edge. Proc. Example Embedded AI Workshop, 2024.